KUE Minilik I Science Shared Campus

EMER-BRACE:A Smart Emergency Bracelet For Instant Emergency Alerts

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Abstract

Every year, around 5 million people from stroke patients die due to delay in the first one, golden hour. 71% of our world's death is because of chronic illnesses, with over 85% occurring in low- and middle-income countries in which many are linked to delayed medical care or unmonitored health episodes.

Thus the team designed a wearable medical alert bracelet for high-risk patients such as pregnant mothers, individuals with epilepsy, cardiac and stroke risk patients, and other chronic illness cases. The system detects events of emergency, such as falls and abnormal heart rate, oxygen concentration, and body temperature, and reports to the patient's assigned hospital and caretaker through GSM-based SMS with attached GPS location, reducing preventable deaths from delayed clinical intervention.

The system integrats sensors such as Max30100 for Pulse and SpO_2 monitoring, GY906 for temperature sensing, MPU6050 for motion and fall detection, and GSM/GPS modules for communication and location tracking together with manual SOS Button.

Unlike traditional alert systems that only notify nearby contacts or require smart phone connectivity, EMER-BRACE directly links them to hospitals ensuring faster and more reliable medical response. With this smart bracelet's features, namely Registered patients with a Device automatically linked to hospitals, Immediate alert using SMS + GPS coordinates sent to hospital & caregivers to provide rapid response, continuous monitoring of Pulse & temperature which provides early warning for anomalies, and manual SOS Button that allows patient to trigger alert if symptoms appear, we can reduce the significance by a big rate.

This device offers people a low-cost, life saving technology advancing SDG 3(Good Health and Well-being) and SDG 9(Industry, Innovation and Infrastructure) to collaborate and create a more healthier and safer community.

Key Words: Emergency Response, Health Monitoring, GSM based SMS, Fall detection, Wearable bracelet

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